Tensor Networks Methods

Exercise Set 1: Symmetries of 1D Spin Chains

Exercise 1: Translational Symmetry & Single-Magnon Excitations

Consider a 1D spin-$\frac{1}{2}$ chain of N sites with periodic boundary conditions (PBC), described by the translation operator T:

$$T|s_1, s_2, \ldots, s_N| = |s_N, s_1, \ldots, s_{N-1}|, \quad \text{where } s_j \in \{\uparrow, \downarrow\}. \tag{1}$$

Questions

1.1 Spectrum of T: Show that $T^N = \mathbb{I}$ and derive the allowable eigenvalues $\lambda_k = e^{-ik}$ of T, expressing k in terms of discrete quantum numbers.

1.2 Commutation with H_{XXX}: For the periodic XXX Hamiltonian $H_{XXX} = J\sum_{j=1}^{N} \mathbf{S}_j \cdot \mathbf{S}_{j+1}$ (with $\mathbf{S}_{N+1} \equiv \mathbf{S}_1$), prove that $[H_{XXX}, T] = 0$. What is the physical consequence?

1.3 Single-Magnon State: Let $|\mathbf{0}| = |\uparrow\uparrow \ldots \uparrow|$ be the ferromagnetic reference state.

(a) Verify that $H_{XXX}|\mathbf{0}| = E_0|\mathbf{0}|$ and find E_0.

(b) Define single-flip momentum states $|k| = \frac{1}{\sqrt{N}}\sum_{j=1}^{N} e^{ikj}|j|$, where $|j| = S_j^-|\mathbf{0}|$. Show that $T|k| = e^{-ik}|k|$.

(c) Derive the dispersion relation $E(k) - E_0$ for a single magnon excitation.

Exercise 2: Global $SU(2)$ Symmetry of the XXX Chain

Define total spin operators $S^\alpha = \sum_{j=1}^{N} S_j^\alpha$ for $\alpha \in \{x, y, z\}$, and total spin squared $\mathbf{S}^2 = (S^x)^2 + (S^y)^2 + (S^z)^2$.

Questions

2.1 Commutation Relations: Prove that $[H_{XXX}, S^\alpha] = 0$ for all $\alpha \in \{x, y, z\}$, and show that $[H_{XXX}, \mathbf{S}^2] = 0$.

2.2 Multiplet Structure: Explain why energy levels must form $SU(2)$ representations of dimension $(2S+1)$. Prove that if $|\psi|$ is an eigenstate with energy E, then $S^\pm|\psi|$ are also eigenstates with the exact same energy E.

2.3 Dimension Reduction: How does $SU(2)$ symmetry simplify the exact diagonalization of H_{XXX}?

Exercise 3: Symmetry Reduction in the XXZ Chain

The anisotropic XXZ spin chain Hamiltonian is:

$$H_{XXZ} = J \sum_{j=1}^{N} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right), \quad \Delta \neq 1. \tag{2}$$

Questions

3.1 Continuous Symmetry Reduction: Prove that for $\Delta \neq 1$, $[H_{XXZ}, S^z] = 0$, but $[H_{XXZ}, S^x] \neq 0$. What is the remaining continuous symmetry group?

3.2 Spin-Flip Parity (Discrete Symmetry): Define $\mathcal{P} = \prod_{j=1}^{N} \sigma_j^x$, where $\sigma_j^x = 2S_j^x$.

(a) Show that $\mathcal{P} S_j^z \mathcal{P}^\dagger = -S_j^z$ and $\mathcal{P} S^z \mathcal{P}^\dagger = -S^z$.

(b) Prove that $[H_{XXZ}, \mathcal{P}] = 0$.

(c) Does $\mathcal{P}$ commute with S^z? What does this imply for simultaneous quantum numbers?

3.3 Spatial Inversion Parity: Let I be spatial inversion ($I S_j^\alpha I^\dagger = S_{N-j+1}^\alpha$). Show that $[H_{XXZ}, I] = 0$ and derive $ITI^\dagger = T^{-1}$. What does this imply for $E(k)$ versus $E(-k)$?

Exercise 4: The Spin to Hard-Core Boson Mapping

Consider a 1D lattice with spin-1/2 operators $\mathbf{S}_j = \frac{1}{2}\boldsymbol{\sigma}_j$ at each site $j \in \{1, \ldots, N\}$. Define the boson creation and annihilation operators $a_j^\dagger, a_j$ via:

$$a_j^\dagger = S_j^+ = S_j^x + iS_j^y, \quad a_j = S_j^- = S_j^x - iS_j^y. \tag{3}$$

Questions

4.1 On-site Algebra: Show that a_j and $a_j^\dagger$ satisfy on-site anti-commutation relations:

$$\{a_j, a_j^\dagger\} = 1, \quad (a_j^\dagger)^2 = 0, \quad (a_j)^2 = 0. \tag{4}$$

What is the physical interpretation of $(a_j^\dagger)^2 = 0$?

4.2 Inter-site Algebra: Show that for different sites ($j \neq k$), a_j and $a_k^\dagger$ commute:

$$[a_j, a_k^\dagger] = 0, \quad [a_j, a_k] = 0 \quad (j \neq k). \tag{5}$$

Explain why these operators are called **hard-core bosons**.

4.3 Hamiltonian Mapping: Map the XXZ Hamiltonian with a transverse magnetic field h:

$$H = -J \sum_{j=1}^{N-1} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right) - h \sum_{j=1}^{N} S_j^z, \tag{6}$$

into a system of hard-core bosons. Express the parameters of the boson model (hopping amplitude t, nearest-neighbor interaction V, chemical potential μ) in terms of J, Δ, and h.

Exercise 5: The Jordan–Wigner Transformation

Hard-core bosons commute at different sites, whereas true physical fermions must anti-commute across all sites. To map hard-core bosons to spineless fermions $c_j, c_j^\dagger$, we attach a non-local phase factor (the Jordan–Wigner string).

Define:

$$c_j = \exp\left(i\pi \sum_{l=1}^{j-1} n_l\right) a_j, \quad c_j^\dagger = a_j^\dagger \exp\left(-i\pi \sum_{l=1}^{j-1} n_l\right), \tag{7}$$

where $n_l = a_l^\dagger a_l = c_l^\dagger c_l$.

Questions

5.1 Fermionic Anti-commutation: Show that $K_j = \exp\left(i\pi \sum_{l=1}^{j-1} n_l\right) = \prod_{l=1}^{j-1}(-\sigma_l^z)$.

5.2 Prove that the operators $c_j, c_k^\dagger$ satisfy canonical fermionic anti-commutation relations:

$$\{c_j, c_k^\dagger\} = \delta_{jk}, \quad \{c_j, c_k\} = 0. \tag{8}$$

5.3 Local Bilinear Invariance: Prove that nearest-neighbor hopping of hard-core bosons maps directly to local fermionic hopping:

$$a_j^\dagger a_{j+1} = c_j^\dagger c_{j+1}. \tag{9}$$

Explain why the non-local string cancels out for nearest neighbors.

Exercise 6: Exact Solution of the Transverse Field Ising Model (TFIM)

Consider the 1D Transverse Field Ising Model on an open chain:

$$H_{\text{TFIM}} = -J \sum_{j=1}^{N-1} \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^{N} \sigma_j^z. \tag{10}$$

Questions

6.1 Jordan–Wigner Representation: Express σ_j^x and σ_j^z in terms of fermionic operators $c_j, c_j^\dagger$. Show that H_{TFIM} maps to a quadratic quadratic fermionic Hamiltonian (Kitaev chain):

$$H_{\text{TFIM}} = -J \sum_{j=1}^{N-1} \left(c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger + \text{h.c.}\right) - 2h \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2}\right). \tag{11}$$

6.2 Fourier Transform (Periodic Chain): Assuming periodic boundary conditions ($c_{N+1} \equiv c_1$) and introducing momentum operators $c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k$, write H_{TFIM} in momentum space as:

$$H_{\text{TFIM}} = \sum_{k>0} \begin{pmatrix} c_k^\dagger & c_{-k} \end{pmatrix} \mathcal{H}(k) \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}, \tag{12}$$

and find the 2×2 matrix $\mathcal{H}(k)$.

6.3 Bogoliubov Spectrum & Quantum Phase Transition:

(a) Diagonalize $\mathcal{H}(k)$ using a Bogoliubov transformation:

$$\eta_k = u_k c_k - i v_k c_{-k}^\dagger, \tag{13}$$

and derive the excitation spectrum $\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh\cos k}$.

(b) Find the critical field h_c where the excitation gap vanishes at $k = 0$. What physical transition occurs at $h = h_c$?

Solutions

Detailed Solution to Exercise 1

Solution 1.1: Applying T N times shifts all spin labels by N positions, returning the state to itself:

$$T^N|s_1, s_2, \ldots, s_N| = |s_1, s_2, \ldots, s_N| \implies T^N = \mathbb{I}. \tag{14}$$

If λ is an eigenvalue of T, then $\lambda^N = 1$. Writing $\lambda = e^{-ik}$, we get $e^{-ikN} = 1$, which requires:

$$kN = 2\pi m \implies k = \frac{2\pi m}{N}, \quad m \in \{0, 1, \ldots, N-1\}. \tag{15}$$

The quantity k is the crystal momentum defined within the first Brillouin zone $[0, 2\pi)$.

Solution 1.2: The operator T maps local spin operators according to $TS_j^\alpha T^\dagger = S_{j+1}^\alpha$. Applying T to H_{XXX}:

$$TH_{XXX}T^\dagger = J\sum_{j=1}^{N} T(\mathbf{S}_j \cdot \mathbf{S}_{j+1})T^\dagger = J\sum_{j=1}^{N} \mathbf{S}_{j+1} \cdot \mathbf{S}_{j+2} = H_{XXX}, \tag{16}$$

where periodic boundary conditions ensure that site $N+1 \equiv 1$ and $N+2 \equiv 2$. Since $TH_{XXX}T^\dagger = H_{XXX}$, it follows that $[H_{XXX}, T] = 0$. **Physical consequence:** H_{XXX} and T share a common eigenbasis. Energy eigenstates can be labeled by their crystal momentum k.

Solution 1.3(a): Rewrite $\mathbf{S}_j \cdot \mathbf{S}_{j+1} = \frac{1}{2}(S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+) + S_j^z S_{j+1}^z$. Acting on $|\mathbf{0}|$:

- $S_j^+|\mathbf{0}| = 0$, so the flip terms yield zero: $(S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+)|\mathbf{0}| = 0$.
- $S_j^z|\uparrow\rangle = \frac{1}{2}|\uparrow\rangle$, so $S_j^z S_{j+1}^z|\mathbf{0}| = \frac{1}{4}|\mathbf{0}|$.

Summing over N bonds gives $H_{XXX}|\mathbf{0}| = JN\left(\frac{1}{4}\right)|\mathbf{0}| = \frac{JN}{4}|\mathbf{0}|$. Thus $E_0 = \frac{JN}{4}$.

Solution 1.3(b): Note that $T|j| = |j+1|$ with $|N+1| \equiv |1|$. Applying T to $|k|$:

$$T|k| = \frac{1}{\sqrt{N}}\sum_{j=1}^{N} e^{ikj} T|j| = \frac{1}{\sqrt{N}}\sum_{j=1}^{N} e^{ikj}|j+1|. \tag{17}$$

Shift the summation index $j' = j + 1$:

$$T|k| = \frac{1}{\sqrt{N}}\sum_{j'=1}^{N} e^{ik(j'-1)}|j'| = e^{-ik}\left(\frac{1}{\sqrt{N}}\sum_{j'=1}^{N} e^{ikj'}|j'|\right) = e^{-ik}|k|. \tag{18}$$

Solution 1.3(c): Calculate $H_{XXX}|j|$: The $S_l^z S_{l+1}^z$ term gives eigenvalue $\frac{1}{4}$ for $(N-2)$ un-flipped bonds and $-\frac{1}{4}$ for the 2 bonds touching site j. Contribution: $\frac{1}{4}(N-2) - \frac{1}{4}(2) = \frac{N-4}{4}$. The transverse term $\frac{1}{2}(S_l^+ S_{l+1}^- + S_l^- S_{l+1}^+)$ flips the single down spin to neighboring sites: $\frac{1}{2}|j-1| + \frac{1}{2}|j+1|$. Thus:

$$H_{XXX}|j| = J\left[\frac{N-4}{4}|j| + \frac{1}{2}(|j-1| + |j+1|)\right]. \tag{19}$$

Now apply H_{XXX} to $|k|$:

$$\begin{aligned} H_{XXX}|k| &= \frac{J}{\sqrt{N}} \sum_{j=1}^{N} e^{ikj} \left[\frac{N-4}{4}|j| + \frac{1}{2}(|j-1| + |j+1|) \right] \\ &= J \left[\frac{N-4}{4} + \frac{1}{2}(e^{ik} + e^{-ik}) \right] |k| \\ &= \left[\frac{JN}{4} - J + J\cos k \right] |k| = [E_0 - J(1-\cos k)]\,|k|. \end{aligned} \tag{20}$$

The single-magnon excitation energy above ground state E_0 is:

$$\Delta E(k) = E(k) - E_0 = J(1-\cos k) = 2J\sin^2\left(\frac{k}{2}\right). \tag{21}$$

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Detailed Solution to Exercise 2

Solution 2.1: Consider $[H_{XXX}, S^z]$. Express $H_{XXX} = J\sum_j \mathbf{S}_j \cdot \mathbf{S}_{j+1}$. Using standard spin commutator $[S_j^\alpha, S_k^\beta] = i\epsilon_{\alpha\beta\gamma}\delta_{jk}S_j^\gamma$:

$$\begin{aligned} [S_j^x S_{j+1}^x + S_j^y S_{j+1}^y, S_j^z + S_{j+1}^z] &= [S_j^x, S_j^z]S_{j+1}^x + S_j^x[S_{j+1}^x, S_{j+1}^z] + [S_j^y, S_j^z]S_{j+1}^y + S_j^y[S_{j+1}^y, S_{j+1}^z] \\ &= -iS_j^y S_{j+1}^x - iS_j^x S_{j+1}^y + iS_j^x S_{j+1}^y + iS_j^y S_{j+1}^x = 0. \end{aligned} \tag{22}$$

Because H_{XXX} is fully isotropic under coordinate rotations, the same algebraic steps hold for S^x and S^y. Hence $[H_{XXX}, S^\alpha] = 0$. Since $\mathbf{S}^2 = \sum_\alpha (S^\alpha)^2$, we have:

$$[H_{XXX}, \mathbf{S}^2] = \sum_{\alpha=x,y,z} ([H_{XXX}, S^\alpha]S^\alpha + S^\alpha[H_{XXX}, S^\alpha]) = 0. \tag{23}$$

Solution 2.2: Because H_{XXX} commutes with S^z and $S^\pm = S^x \pm iS^y$:

$$H_{XXX}(S^\pm|\psi|) = S^\pm H_{XXX}|\psi| = S^\pm(E|\psi|) = E(S^\pm|\psi|). \tag{24}$$

Thus, $S^\pm|\psi|$ are energy eigenstates with eigenvalue E. Repeating S^- constructs all $2S+1$ states of a spin-S multiplet $|S, M\rangle$ (with $M = -S, \ldots, S$), all having identical energy E.

Solution 2.3: By $SU(2)$ symmetry, all states in a multiplet are fully determined by the **highest-weight state** $|\psi_{\text{HW}}\rangle$, defined by:

$$S^+|\psi_{\text{HW}}\rangle = 0, \quad S^z|\psi_{\text{HW}}\rangle = S|\psi_{\text{HW}}\rangle. \tag{25}$$

Instead of diagonalizing H_{XXX} across the full 2^N-dimensional Hilbert space, one can restrict search exclusively to the much smaller highest-weight subspace for each S^z sector.

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Detailed Solution to Exercise 3

Solution 3.1: Decompose $H_{XXZ} = H_{XX} + \Delta H_Z$, where $H_{XX} = J\sum_j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y)$ and $H_Z = J\sum_j S_j^z S_{j+1}^z$. $[H_Z, S^z] = 0$ and $[H_{XX}, S^z] = 0$, so $[H_{XXZ}, S^z] = 0$. However, $[H_Z, S^x] \neq 0$ because $[S_j^z S_{j+1}^z, S_j^x + S_{j+1}^x] = iS_j^y S_{j+1}^z + iS_j^z S_{j+1}^y \neq 0$. Thus, for $\Delta \neq 1$, $SU(2)$ is explicitly broken down to $U(1)$ continuous spin rotations around the z-axis.

Solution 3.2(a): Since $\sigma^x\sigma^z\sigma^x = -\sigma^z$, we have $\mathcal{P}S_j^z\mathcal{P}^\dagger = -S_j^z$. Summing over j:

$$\mathcal{P}S^z\mathcal{P}^\dagger = \mathcal{P}\left(\sum_{j=1}^{N} S_j^z\right)\mathcal{P}^\dagger = \sum_{j=1}^{N}(-S_j^z) = -S^z. \tag{26}$$

Solution 3.2(b): Under $\mathcal{P}$: $S_j^x \to S_j^x$, $S_j^y \to -S_j^y$, and $S_j^z \to -S_j^z$. Evaluating H_{XXZ} transformation:

- $S_j^x S_{j+1}^x \to (+S_j^x)(+S_{j+1}^x) = S_j^x S_{j+1}^x$
- $S_j^y S_{j+1}^y \to (-S_j^y)(-S_{j+1}^y) = S_j^y S_{j+1}^y$
- $S_j^z S_{j+1}^z \to (-S_j^z)(-S_{j+1}^z) = S_j^z S_{j+1}^z$

Every term in H_{XXZ} is invariant under $\mathcal{P}$. Hence $[H_{XXZ}, \mathcal{P}] = 0$.

Solution 3.2(c): Calculate $\mathcal{P}S^z + S^z\mathcal{P} = 0 \implies [\mathcal{P}, S^z] = 2\mathcal{P}S^z \neq 0$ (unless $S^z = 0$). Since $\mathcal{P}$ and S^z do not commute, energy eigenstates cannot generally have simultaneous well-defined S^z and spin-flip parity values, except in the $S^z = 0$ sector where $\mathcal{P}$ maps the subspace onto itself.

Solution 3.3: Inversion I maps $j \to N - j + 1$. Since H_{XXZ} contains sums over all nearest-neighbor bonds $(j, j+1)$, inversion simply reverses the sum order, leaving H_{XXZ} invariant: $[H_{XXZ}, I] = 0$. For the translation operator T: T advances site indices $j \to j + 1$. $ITI^\dagger$ sends $j \to N - j + 1 \to N - j \equiv j - 1$, which is the backward shift T^{-1}. If $|k\rangle$ has momentum k ($T|k\rangle = e^{-ik}|k\rangle$), then:

$$T(I|k\rangle) = I(I^\dagger T I)|k\rangle = IT^{-1}|k\rangle = I(e^{ik}|k\rangle) = e^{ik}(I|k\rangle). \tag{27}$$

Thus, $I|k\rangle$ is an eigenstate with momentum $-k$. Since $[H_{XXZ}, I] = 0$, $|k\rangle$ and $I|k\rangle$ have identical energy:

$$E(k) = E(-k). \tag{28}$$

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Detailed Solution to Exercise 4

Solution 4.1: Using standard spin relations $S_j^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $S_j^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ in the site basis $\{|\uparrow\rangle, |\downarrow\rangle\}$:

$$\{a_j, a_j^\dagger\} = S_j^- S_j^+ + S_j^+ S_j^- = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \mathbb{I}. \tag{29}$$

Furthermore:

$$(a_j^\dagger)^2 = (S_j^+)^2 = 0, \quad (a_j)^2 = (S_j^-)^2 = 0. \tag{30}$$

Physical interpretation: The condition $(a_j^\dagger)^2 = 0$ enforces the constraint that a single lattice site cannot hold more than one boson (infinite on-site repulsion $U \to \infty$).

Solution 4.2: Spin operators at different sites act on different factors of the tensor product Hilbert space $\mathcal{H} = \bigotimes_{j=1}^{N} \mathcal{H}_j$. Thus, for $j \neq k$, all local spin operators commute:

$$[S_j^\alpha, S_k^\beta] = 0 \quad (\forall \alpha, \beta \in \{x, y, z\}, j \neq k). \tag{31}$$

Consequently, $[a_j, a_k^\dagger] = 0$ for $j \neq k$. These particles commute like standard bosons across different sites, but satisfy Pauli exclusion on the same site — hence the term **hard-core bosons**.

Solution 4.3: Express spin operators in terms of $a_j, a_j^\dagger$, and the number operator $n_j = a_j^\dagger a_j = S_j^+ S_j^- = S_j^z + \frac{1}{2}$:

$$S_j^+ = a_j^\dagger, \quad S_j^- = a_j, \quad S_j^z = n_j - \frac{1}{2}. \tag{32}$$

Now transform the terms in H:

(i) Transverse term: $S_j^x S_{j+1}^x + S_j^y S_{j+1}^y = \frac{1}{2}(S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+) = \frac{1}{2}(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j)$.

(ii) S_z interaction term: $S_j^z S_{j+1}^z = \left(n_j - \frac{1}{2}\right)\left(n_{j+1} - \frac{1}{2}\right) = n_j n_{j+1} - \frac{1}{2}(n_j + n_{j+1}) + \frac{1}{4}$.

(iii) Magnetic field term: $-hS_j^z = -hn_j + \frac{h}{2}$.

Substituting these into H:

$$H = -\frac{J}{2} \sum_{j=1}^{N-1} (a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j) - J\Delta \sum_{j=1}^{N-1} n_j n_{j+1} - \mu \sum_{j=1}^{N} n_j + E_{\text{const}}, \tag{33}$$

where:

- Hopping amplitude: $t = \frac{J}{2}$.
- Nearest-neighbor interaction: $V = -J\Delta$ (attraction for $\Delta > 0$, repulsion for $\Delta < 0$).
- Chemical potential: $\mu = h - J\Delta$ (for bulk sites).
- Constant energy shift: $E_{\text{const}} = \frac{Nh}{2} - \frac{(N-1)J\Delta}{4}$.

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Detailed Solution to Exercise 5

Solution 5.1: Since n_l has eigenvalues 0 and 1, $e^{i\pi n_l} = 1 - 2n_l = 1 - 2\left(S_l^z + \frac{1}{2}\right) = -2S_l^z = -\sigma_l^z$. Therefore:

$$K_j = \prod_{l=1}^{j-1} e^{i\pi n_l} = \prod_{l=1}^{j-1} (-\sigma_l^z). \tag{34}$$

Solution 5.2: **Case 1: $j = k$.** Since K_j depends only on sites $l < j$, K_j commutes with both a_j and $a_j^\dagger$. Furthermore, $K_j^\dagger K_j = \mathbb{I}$.

$$\{c_j, c_j^\dagger\} = \{K_j a_j, a_j^\dagger K_j^\dagger\} = K_j \{a_j, a_j^\dagger\} K_j^\dagger = K_j \mathbb{I} K_j^\dagger = \mathbb{I}. \tag{35}$$

Case 2: $j < k$. Write $c_k = K_k a_k = K_j e^{i\pi n_j} \left(\prod_{l=j+1}^{k-1} e^{i\pi n_l}\right) a_k$. Compute $c_j c_k^\dagger$: The operator $c_j = K_j a_j$ contains a_j. When moving a_j past $c_k^\dagger$, note that $c_k^\dagger$ contains $e^{-i\pi n_j} = (1 - 2n_j)$. We evaluate $a_j(1 - 2n_j)$:

$$a_j(1 - 2a_j^\dagger a_j) = a_j - 2a_j a_j^\dagger a_j = a_j - 2(1 - a_j^\dagger a_j)a_j = -a_j = -(1 - 2n_j)a_j. \tag{36}$$

Thus, a_j anti-commutes with $e^{-i\pi n_j}$. Since a_j commutes with all other site operators in $c_k^\dagger$, we find:

$$c_j c_k^\dagger = -c_k^\dagger c_j \implies \{c_j, c_k^\dagger\} = 0 \quad (j \neq k). \tag{37}$$

Identical steps show $\{c_j, c_k\} = 0$. Hence, $c_j, c_k^\dagger$ are genuine fermionic operators!

Solution 5.3: Compute $c_j^\dagger c_{j+1}$:

$$c_j^\dagger c_{j+1} = \left(a_j^\dagger K_j^\dagger\right)\left(K_{j+1} a_{j+1}\right). \tag{38}$$

Notice $K_{j+1} = K_j e^{i\pi n_j}$. Thus:

$$c_j^\dagger c_{j+1} = a_j^\dagger K_j^\dagger K_j e^{i\pi n_j} a_{j+1} = a_j^\dagger e^{i\pi n_j} a_{j+1}. \tag{39}$$

Now evaluate $a_j^\dagger e^{i\pi n_j} = a_j^\dagger(1 - 2n_j)$:

$$a_j^\dagger(1 - 2a_j^\dagger a_j) = a_j^\dagger - 2(a_j^\dagger)^2 a_j = a_j^\dagger - 0 = a_j^\dagger. \tag{40}$$

Therefore, the string factor simplifies completely on adjacent sites:

$$c_j^\dagger c_{j+1} = a_j^\dagger a_{j+1}. \tag{41}$$

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Detailed Solution to Exercise 6

Solution 6.1: Using $\sigma_j^x = S_j^+ + S_j^- = a_j^\dagger + a_j$ and $\sigma_j^z = 2c_j^\dagger c_j - 1$:

$$\begin{aligned}
\sigma_j^x \sigma_{j+1}^x &= (c_j^\dagger K_j + K_j c_j)(c_{j+1}^\dagger K_{j+1} + K_{j+1} c_{j+1}) \\
&= (c_j^\dagger - c_j)(c_{j+1}^\dagger + c_{j+1}) \\
&= c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger - c_j c_{j+1}^\dagger - c_j c_{j+1} \\
&= (c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger + \text{h.c.}).
\end{aligned} \tag{42}$$

Inserting this into H_{TFIM} directly yields the quadratic (free-fermion) Kitaev chain Hamiltonian.

Solution 6.2: Substitute $c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k$ into H_{TFIM}:

$$\begin{aligned}
\sum_j c_j^\dagger c_{j+1} &= \sum_k \cos k \, c_k^\dagger c_k, \\
\sum_j c_j^\dagger c_{j+1}^\dagger &= \sum_k e^{ik} c_k^\dagger c_{-k}^\dagger = \frac{1}{2} \sum_k \left(e^{ik} c_k^\dagger c_{-k}^\dagger + e^{-ik} c_{-k}^\dagger c_k^\dagger\right) = i \sum_k \sin k \, c_k^\dagger c_{-k}^\dagger.
\end{aligned} \tag{43}$$

Combining terms into a Nambu basis $\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}$ yields $\mathcal{H}(k)$:

$$\mathcal{H}(k) = \begin{pmatrix} -2J\cos k - 2h & 2iJ\sin k \\ -2iJ\sin k & 2J\cos k + 2h \end{pmatrix} = -2(h + J\cos k)\sigma^z - 2J\sin k \, \sigma^y. \tag{44}$$

Solution 6.3(a): The eigenvalues of $\mathcal{H}(k) = d_z(k)\sigma^z + d_y(k)\sigma^y$ are $\pm E(k) = \pm\sqrt{d_z(k)^2 + d_y(k)^2}$:

$$\begin{aligned}\varepsilon(k) &= \sqrt{4(h + J\cos k)^2 + 4J^2\sin^2 k} \\ &= 2\sqrt{h^2 + 2Jh\cos k + J^2\cos^2 k + J^2\sin^2 k} \\ &= 2\sqrt{J^2 + h^2 + 2Jh\cos k}. \end{aligned} \tag{45}$$

(With the convention $k \to \pi - k$, this takes the standard form $\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh\cos k}$).

Solution 6.3(b): At $k = 0$, the excitation gap is $\Delta_g = \varepsilon(0) = 2|J - h|$. The gap closes ($\Delta_g = 0$) when $h = h_c = J$. **Physical Transition:**

- $h < J$: Ferromagnetic ordered phase with non-zero spontaneous magnetization $\langle\sigma^x\rangle \neq 0$ and two-fold degenerate ground states.
- $h > J$: Paramagnetic disordered phase with a unique gapped ground state polarized along the z-axis.
- $h = h_c$: Quantum Critical Point (QCP) described by a 2D Ising conformal field theory (CFT) with central charge $c = 1/2$.

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